INCH-POUND

MIL-DTL-17587/2(SH)
28 June 2007

DETAIL SPECIFICATION SHEET

# CIRCUIT BREAKERS, ACB, LOW VOLTAGE, ELECTRIC POWER, AIR, REMOVABLE CONSTRUCTION ACB-1600HR, ACB-1600HRC, ACB-1600R, AND ACB-1606HRC

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17587.

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REQUIREMENTS:

TABLE I. Ratings/features of circuit breakers.

| Circuit breaker type | ACB-1600HR (PIN 1) | ACB-1600HR (PIN 2) | ACB-1600HRC | ACB-1600R | ACB-1606HRC |
|---|---|---|---|---|---|
| Rated continuous current (amperes) | 1600 | 1600 | 1600 | 1600 | 1600 |
| Rated maximum voltage (AC volts) /1 | 500 | 500 | 500 | 500 | 500 |
| Frequency (hertz) | 60 | 60 | 60 | 60 | 60 |
| Maximum interrupting rating current (symmetrical amperes) | 85,000 | 85,000 | 85,000 | 50,000 | 85,000 |
| Rated short-time current /2 (amperes) | 85,000 | 85,000 | 85,000 | 50,000 | 85,000 |
| Rated short-time current duration (seconds) | 0.5 | 0.5 | 0.5 | 0.5 | 0.5 |
| Operating mechanism - manual | STD | STD | STD | STD | STD |
| Operating mechanism - electric | STD | STD | STD | STD | STD |
| Maximum weight (lbs) – circuit breaker (removable component) | 485 | 318 | 500 | 408 | 205 |
| Maximum weight (lbs) – cradle (stationary component) | 209 | 180 | 209 | 209 | 278 |
| Number of poles | 3 | 3 | 3 | 3 | 3 |
| Number of overcurrent coils or solid-state sensors | 2 | 3 | 3 | 2 | 3 |
| **Attachments** | | | | | |
| Electric closing mechanism | STD | STD | STD | STD | STD |
| Auxiliary switches (number of contacts) | 8 | 10 | 8 | 8 | 8 |
| Undervoltage lockout device | Optional for all circuit breaker types. | | | | |
| Mechanical position indicator | Optional for all circuit breaker types. | | | | |
| Secondary disconnecting devices | Required for all circuit types, unless otherwise specified. | | | | |
| Cell switch | Optional for all circuit breaker types. | | | | |
| Indicator light | Provisions for mounting required for all circuit breaker types; furnished only when specified. | | | | |
| Hold-closed mechanism | OPT | OPT | OPT | OPT | OPT |
| Undervoltage release device | Optional for all circuit breaker. | | | | |
| Shunt trip device | Required for electrically-operated breakers. | | | | |
| Operations counter | OPT | OPT | OPT | OPT | OPT |



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Table I. Ratings/features of circuit breakers - Continued.

| Circuit breaker type | ACB-1600HR (PIN 1) | ACB-1600HR (PIN 2) | ACB-1600HRC | ACB-1600R | ACB-1606HRC |
|---|---|---|---|---|---|
| Trip on power up (TOPU) device | N/A | N/A | N/A | N/A | OPT |
| Rack-in/rack-out position indicator | STD | STD | STD | STD | STD |

NOTES:

/1 Voltage ratings apply to main current-carrying parts. Operating attachments shall be of voltage specified in 6.2 of MIL-DTL-17587.

/2 Ampere values apply to circuit breakers with overcurrent coils or sensors of maximum current rating within the circuit breaker type.



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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors.

| Type | Coil or sensor rating /1 (amperes) | Long-time pickup setting (amperes) | | | Short-time pickup setting (amperes) | | | | | | | | Maximum instantaneous trip /2 (amperes) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ACB-1600R | 320 | 500 | 640 | 840 | --- | 640 | 840 | 960 | 1280 | 1600 | --- | --- | 16000 |
| | 400 | 640 | 840 | 960 | --- | 800 | 960 | 1280 | 1800 | 2100 | --- | --- | 20000 |
| | 480 | 720 | 960 | 1120 | --- | 960 | 1280 | 1600 | 2100 | 2800 | --- | --- | 24000 |
| | 560 | 840 | 1120 | 1280 | --- | 1120 | 1400 | 1800 | 2400 | 3200 | --- | --- | 28000 |
| | 640 | 960 | 1280 | 1600 | --- | 1280 | 1600 | 2100 | 2800 | 3600 | --- | --- | 32000 |
| | 800 | 1280 | 1600 | 2100 | 1200 | 1600 | 2100 | 2400 | 3200 | 4000 | --- | --- | 36000 |
| | 1000 | 1600 | 2100 | 2400 | --- | 2100 | 2400 | 3200 | 4000 | 4800 | --- | --- | 40000 |
| | 1200 | 1800 | 2400 | 2800 | 1800 | 2400 | 2800 | 3600 | 4800 | 6000 | --- | --- | 40000 |
| | 1400 | 2100 | 2800 | 3600 | --- | 2800 | 3200 | 3600 | 4800 | 6000 | --- | --- | 40000 |
| | 1600 | 2400 | 3200 | 4000 | 2400 | 3200 | 4000 | 4800 | 6000 | 8000 | --- | --- | 40000 |
| ACB-1600HR (PIN 1) | 800 | 1280 | 1600 | 2100 | 1200 | 1600 | 2100 | 2400 | 3200 | 4000 | --- | --- | 64000 |
| | 1000 | 1600 | 2100 | 2400 | --- | 2100 | 2400 | 3200 | 4000 | 4800 | --- | --- | 68000 |
| | 1200 | 1800 | 2400 | 2800 | 1800 | 2400 | 2800 | 3600 | 4800 | 6000 | --- | --- | 68000 |
| | 1400 | 2100 | 2800 | 3600 | --- | 2800 | 3200 | 3600 | 4800 | 6000 | --- | --- | 68000 |
| | 1600 | 2400 | 3200 | 4000 | 2400 | 3200 | 4000 | 4800 | 6000 | 8000 | --- | --- | 68000 |





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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors – Continued.

| Type | Coil or sensor rating /1 (amperes) | Long-time pickup setting (amperes) | | | Short-time pickup setting (amperes) | | | | | | | | Maximum instantaneous trip /2 (amperes) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ACB-1600HR (PIN 2) | 320 | 480 | 640 | 800 | 640 | 800 | 960 | 1280 | 1600 | | | | 80000 |
| | 400 | 600 | 800 | 1000 | 800 | 1000 | 1200 | 1600 | 2000 | | | | 80000 |
| | 480 | 720 | 960 | 1200 | 960 | 1200 | 1440 | 1920 | 2400 | | | | 80000 |
| | 560 | 840 | 1120 | 1400 | 1120 | 1400 | 1680 | 2240 | 2800 | | | | 80000 |
| | 640 | 960 | 1280 | 1600 | 1280 | 1600 | 1920 | 2560 | 3200 | | | | 80000 |
| | 800 | 1200 | 1600 | 2000 | 1600 | 2000 | 2400 | 3200 | 4000 | | | | 80000 |
| | 1000 | 1500 | 2000 | 2500 | 2000 | 2500 | 3000 | 4000 | 5000 | | | | 80000 |
| | 1200 | 1800 | 2400 | 3000 | 2400 | 3000 | 3600 | 4800 | 6000 | | | | 80000 |
| | 1400 | 2100 | 2800 | 3500 | 2800 | 3500 | 4200 | 5600 | 7000 | | | | 80000 |
| | 1600 | 2400 | 3200 | 4000 | 3200 | 4000 | 4800 | 6400 | 8000 | | | | 80000 |



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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors – Continued.

| Type | Coil or sensor rating /1 (amperes) | Long-time pickup setting (amperes) | | | Short-time pickup setting (amperes) | | | | | | | | Maximum instantaneous trip /2 (amperes) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ACB-1600HRC | 800 | 1280 | 1600 | 2100 | 1200 | 1600 | 2100 | 2400 | 3200 | --- | --- | --- | 64000 |
| | 1000 | 1600 | 2100 | 2400 | --- | 2100 | 2400 | 3200 | 4000 | --- | --- | --- | 68000 |
| | 1200 | 1800 | 2400 | 2800 | 1800 | 2400 | 2800 | 3600 | 4800 | --- | --- | --- | 68000 |
| | 1400 | 2100 | 2800 | 3600 | --- | 2800 | 3200 | 3600 | 4800 | --- | --- | --- | 68000 |
| | 1600 | 2400 | 3200 | 4000 | 2400 | 3200 | 4000 | 4800 | 6000 | --- | --- | --- | 68000 |
| ACB-1606HRC | 600 | 900 | 1200 | 1500 | 1200 | 1500 | 1800 | 2400 | 3000 | 3600 | 4200 | 4800 | 32000 |
| | 800 | 1200 | 1600 | 2000 | 1600 | 2000 | 2400 | 3200 | 4000 | 4800 | 5600 | 6400 | 32000 |
| | 1000 | 1500 | 2000 | 2500 | 2000 | 2500 | 3000 | 4000 | 5000 | 6000 | 7000 | 8000 | 32000 |
| | 1200 | 1800 | 2400 | 3000 | 2400 | 3000 | 3600 | 4800 | 6000 | 7200 | 8400 | 9600 | 32000 |
| | 1400 | 2100 | 2800 | 3500 | 2800 | 3500 | 4200 | 5600 | 7000 | 8400 | 9800 | 11200 | 32000 |
| | 1600 | 2400 | 3200 | 4000 | 3200 | 4000 | 4800 | 6400 | 8000 | 9600 | 11200 | 12800 | 32000 |

NOTES:

/1 Continuous current rating.

/2 The instantaneous trip device shall be set at the place of manufacture from five times the coil or sensor rating to the maximum pickup setting specified for mechanical coils and 20 times sensor rating for solid-state sensors with a default at maximum listed rating.



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TABLE III. Electrical endurance and temperature rise test requirements.

| Electrical endurance test: | | | | |
|---|---|---|---|---|
| **Type** | **Total number of close-open cycles** | **Amperes on circuit breaker close** | **Amperes on circuit breaker open** | **Maximum AC power factor (lagging) X/R** |
| ACB-1600HR (PIN 1) | 1,000 | 8,000 | 1,600 | Close: 0.25 maximum; Open: 0.80 maximum |
| ACB-1600HR (PIN 2) | 1,000 | 8,000 | 1,600 | Close: 0.25 maximum; Open: 0.80 maximum |
| ACB-1600HRC | 1,000 | 8,000 | 1,600 | Close: 0.25 maximum; Open: 0.80 maximum |
| ACB-1600R | 1,000 | 8,000 | 1,600 | Close: 0.25 maximum; Open: 0.80 maximum |
| ACB-1606HRC | 4,000 | 8,000 | 1600 | Close: 0.25 maximum; Open: 0.80 maximum |

| Temperature rise test: | | | | | |
|---|---|---|---|---|---|
| **Type** | **Continuous current rating** | **Cable quantity per terminal** | **Cable size /1** | **Bus bar quantity per phase** | **Bus bar size (inches) /2** |
| ACB-1600HR (PIN 1) | 1,600 | 4 | 400 MCM | 2 | 3.0 X ¼ |
| ACB-1600HR (PIN 2) | | | | | |
| ACB-1600HRC | | | | | |
| ACB-1600R | | | | | |
| ACB-1606HRC | | | | | |

NOTE:

/1 Information regarding cable is contained in MIL-DTL-915. If 3-phase cable is used for this test, the cable jacket shall be stripped for a minimum of 2 feet.

/2 Information regarding bus bars is contained in MIL-DTL-16036.



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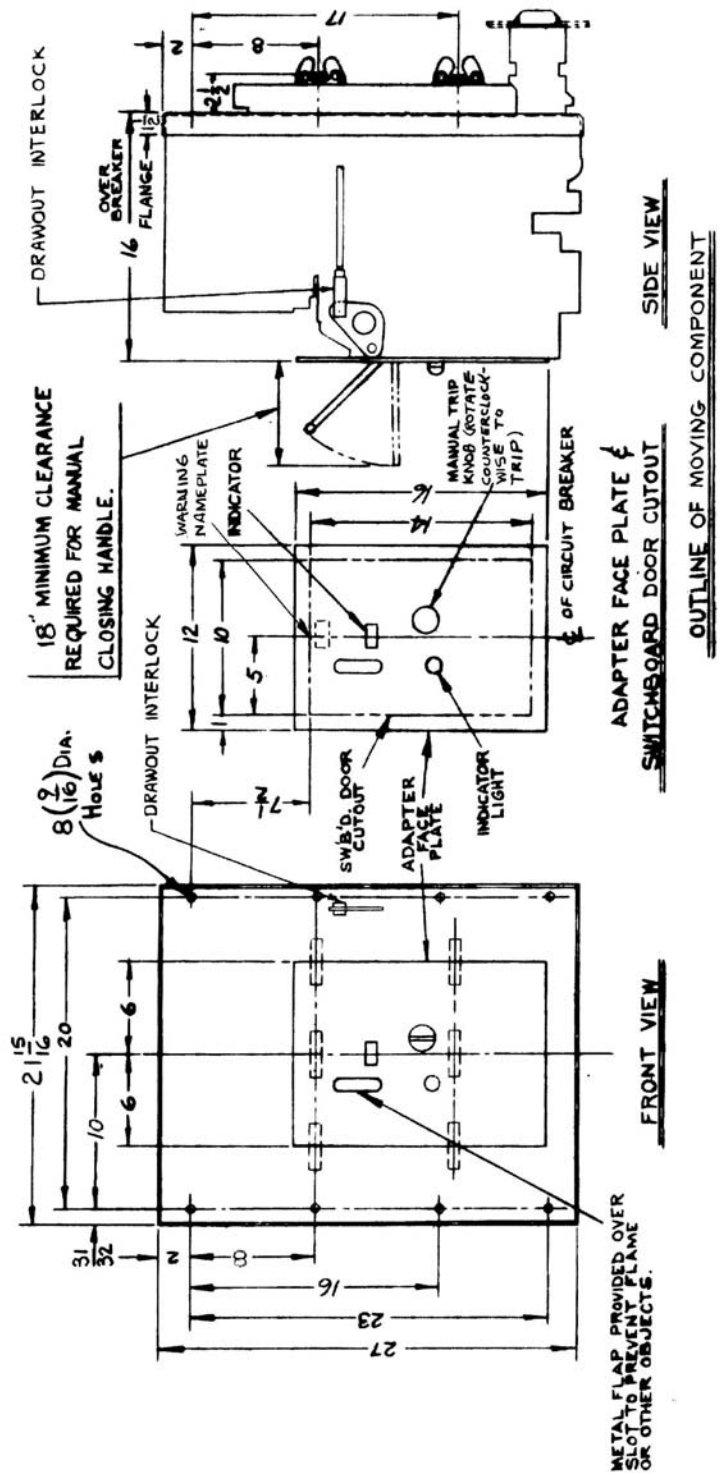



FIGURE 1. Outline and mounting dimensions for circuit breaker types ACB-1600HR (PIN 1) and ACB-1600HRC.



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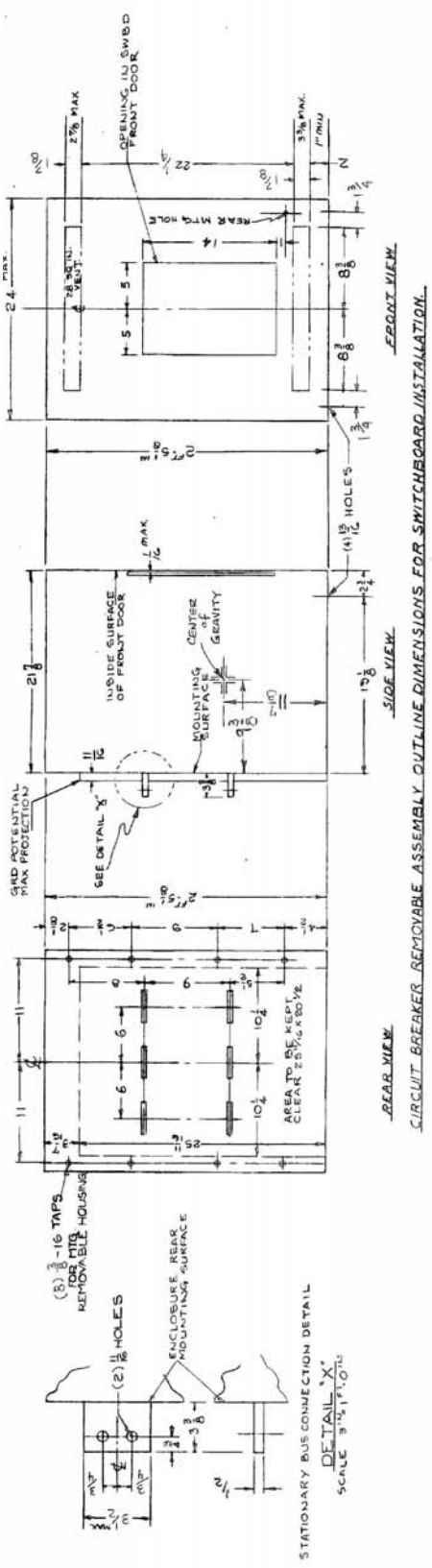



FIGURE 2. Outline and mounting dimensions for circuit breaker types ACB-1600HR (PIN1) and ACB-1600HRC (removable element for).



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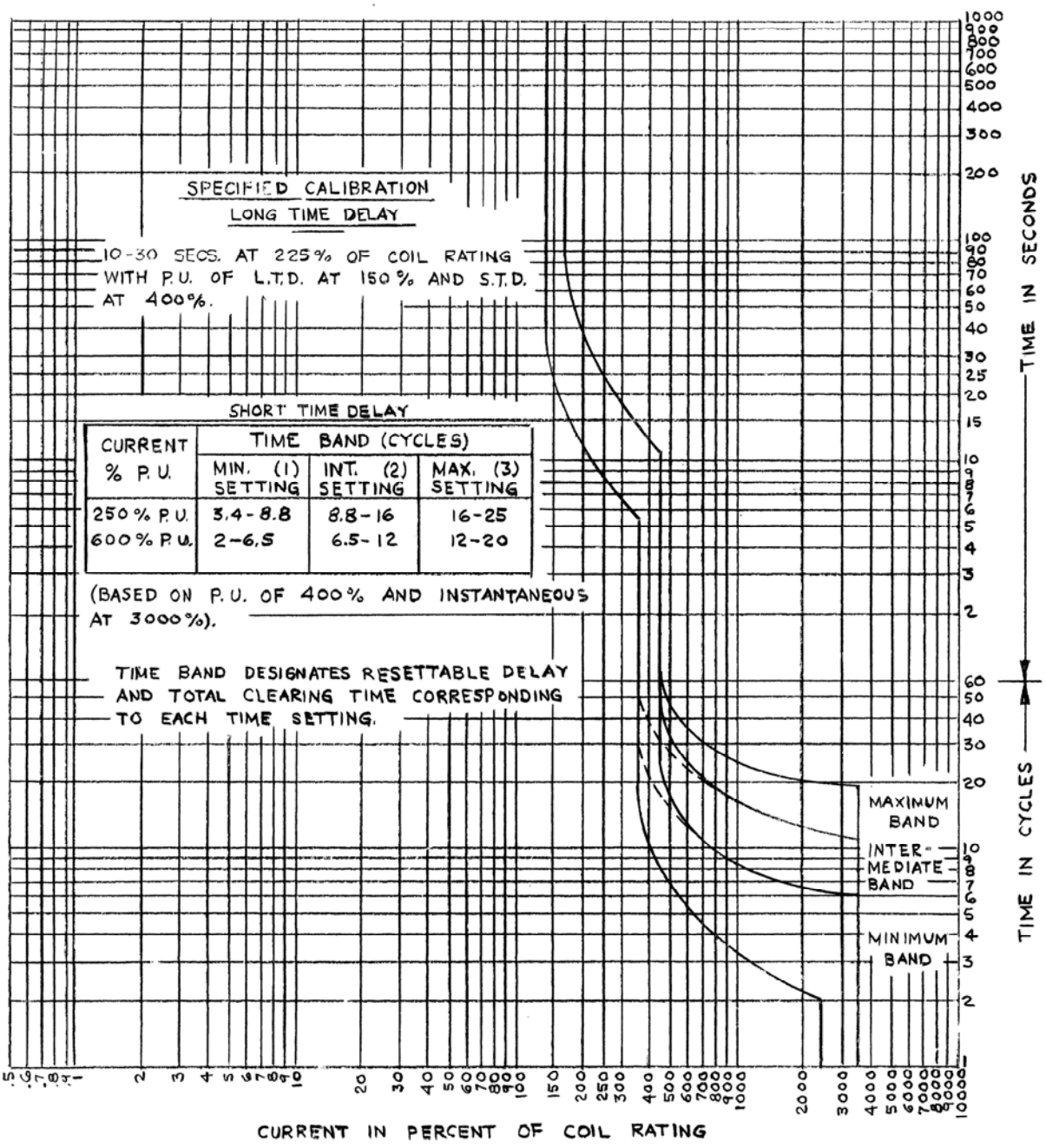



FIGURE 3. Time-current characteristic curve for circuit breaker types ACB-1600HR (PIN 1) and ACB-1600HRC.



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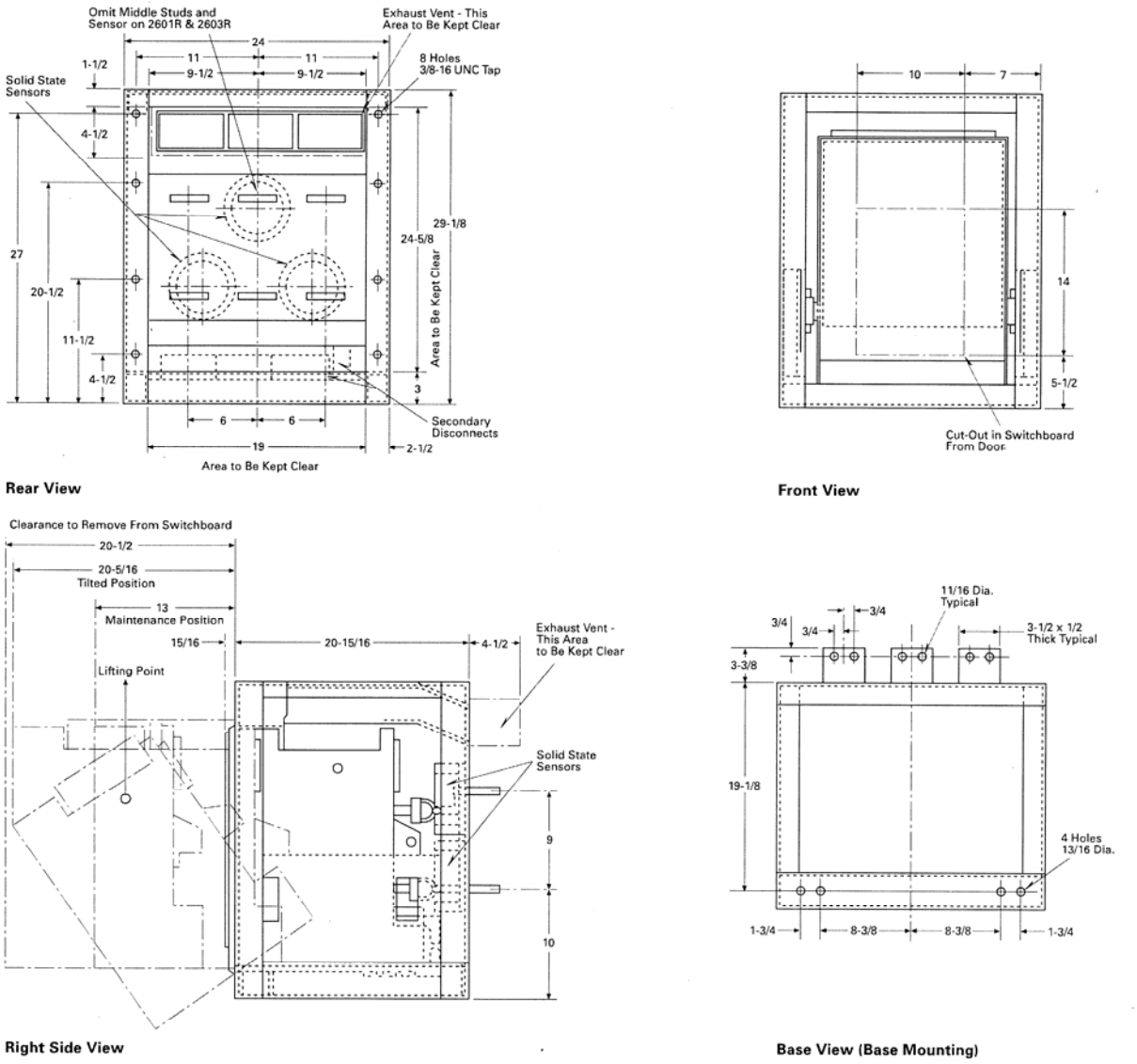



FIGURE 4. Outline and mounting dimensions for circuit breaker type ACB-1600HR (PIN 2).



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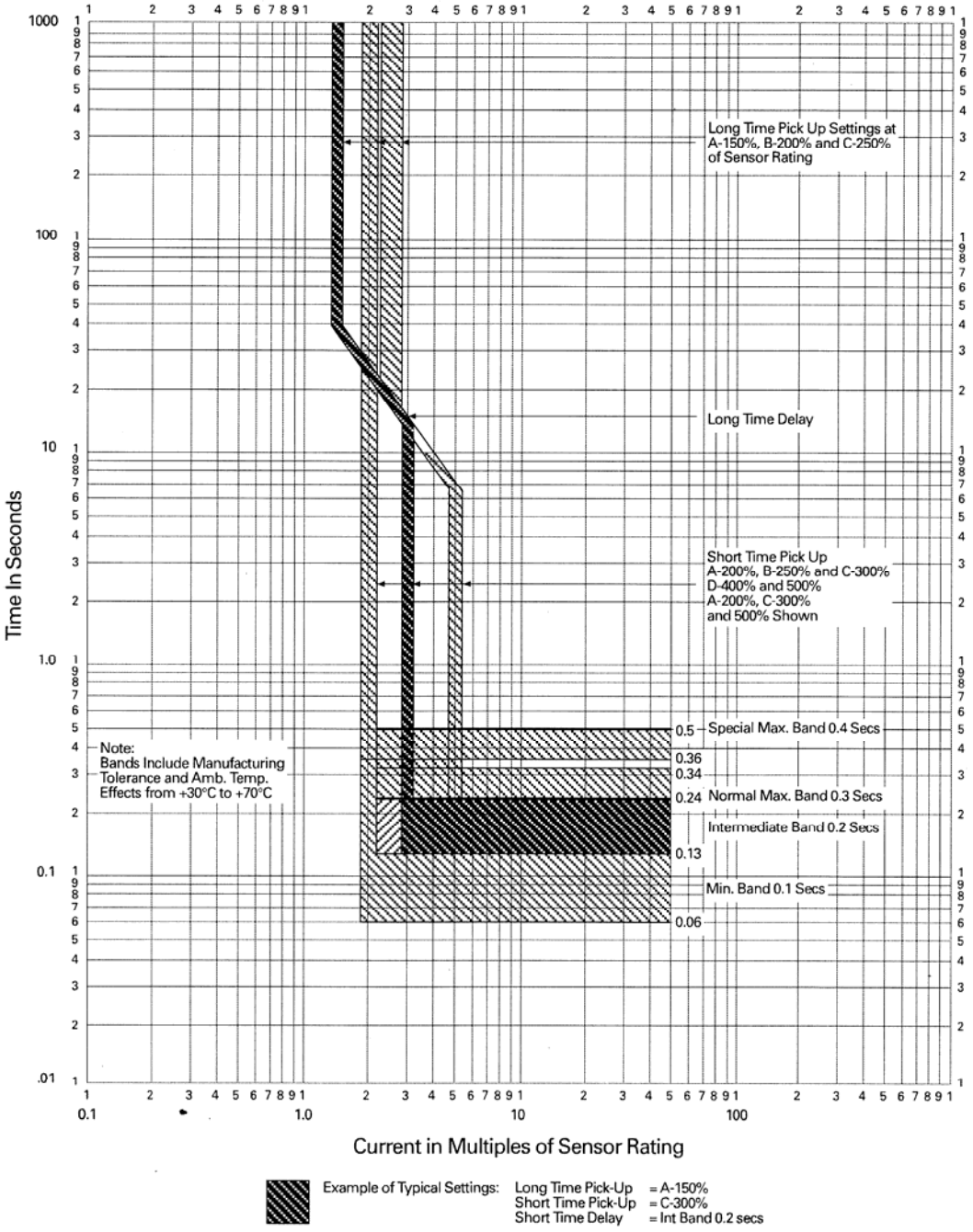



FIGURE 5. Time-current characteristic curve for circuit breaker type ACB-1600HR (PIN 2).



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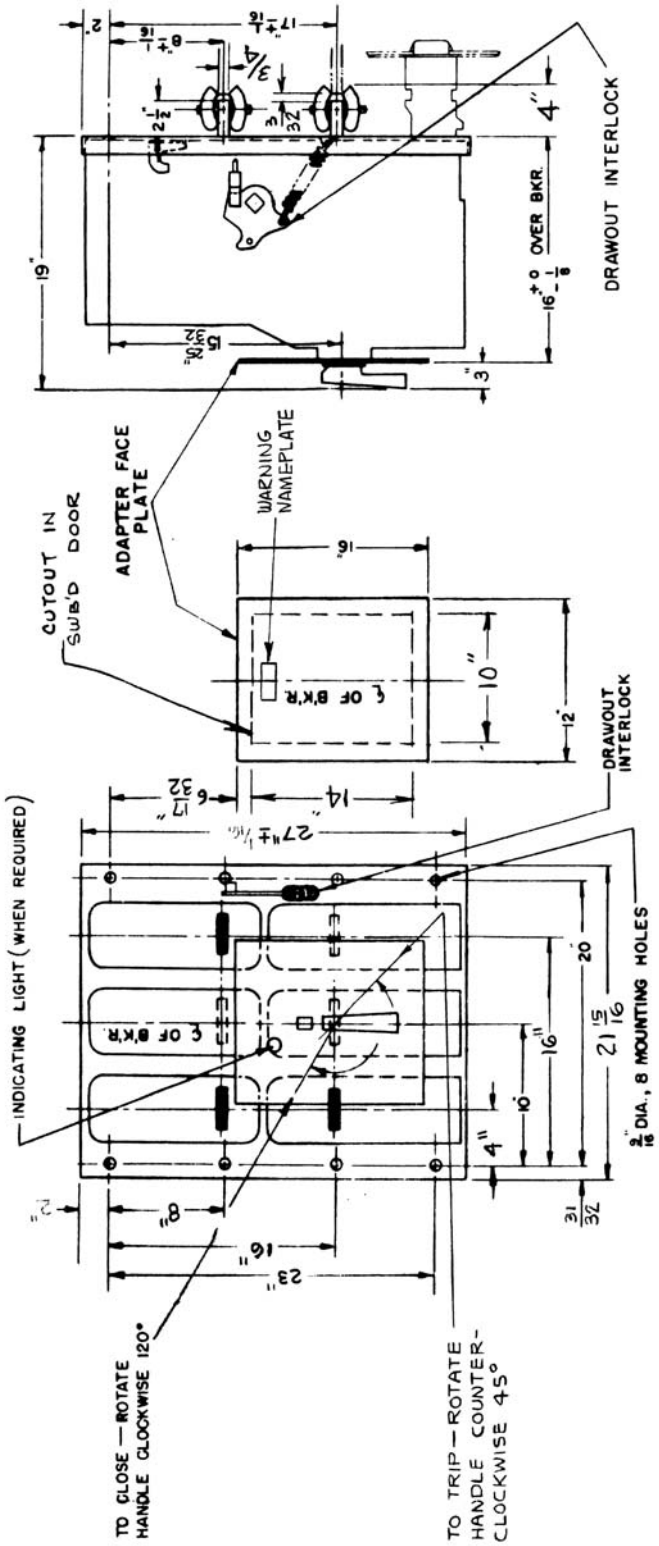



FIGURE 6. Outline and mounting dimensions for circuit breaker type ACB-1600R.



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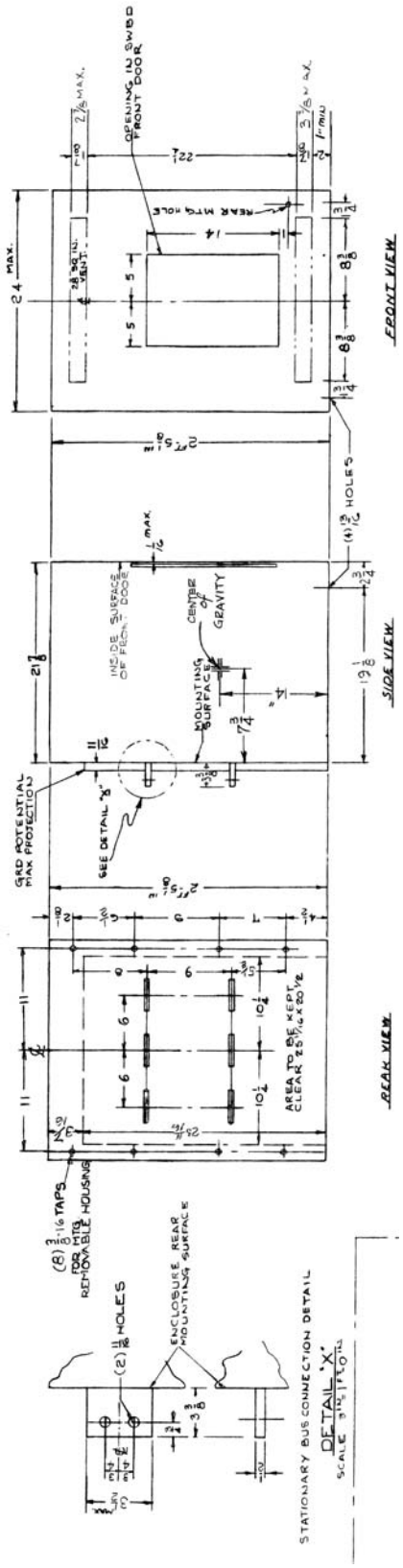



FIGURE 7. Outline and mounting dimensions for circuit breaker type ACB-1600R.



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~SPECIFIED CALIBRATION~

LONG TIME DELAY

10-30 SECONDS AT 225% OF COIL RATING WITH P.U. OF L.T.D. AT 150% AND S.T.D. AT 400%

SHORT TIME DELAY

| CURRENT % P.U. | TIME BAND (CYCLES) | | |
|---|---|---|---|
| | MIN. (1) SETTING | INT. (2) SETTING | MAX. (3) SETTING |
| 250% P.U. | 3.4-8.8 | 8.8-16 | 16-25 |
| 600% P.U. | 2-6.5 | 6.5-12 | 12-20 |

BASED ON P.U. OF 400% AND INSTANTANEOUS AT 3000%

TIME BAND DESIGNATES RESETTABLE DELAY AND TOTAL CLEARING TIME CORRESPONDING TO EACH TIME SETTING.

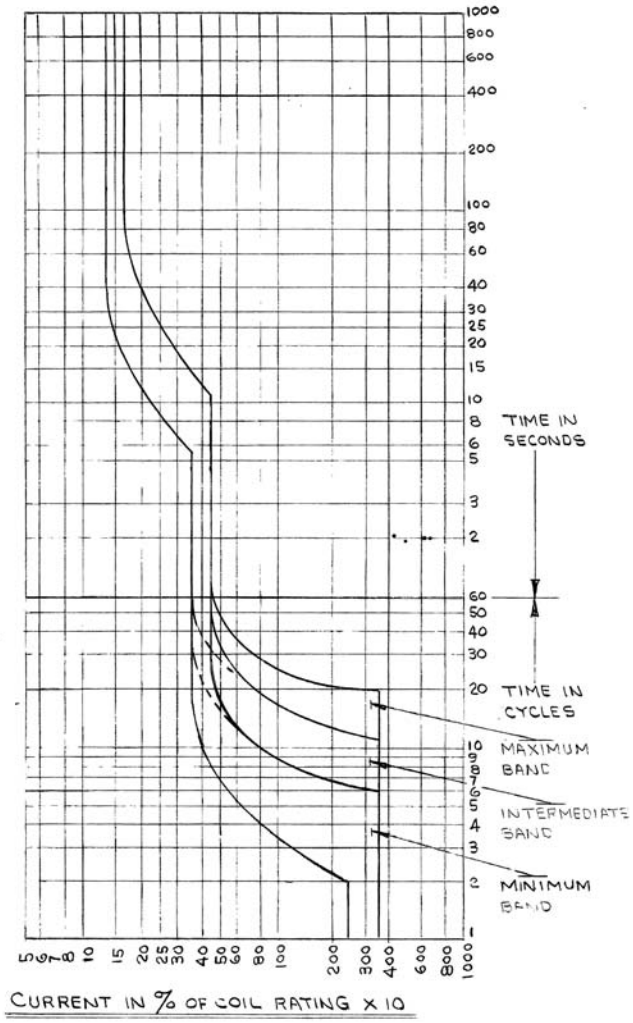



FIGURE 8. Time-current characteristic curve for circuit breaker type ACB-1600R.



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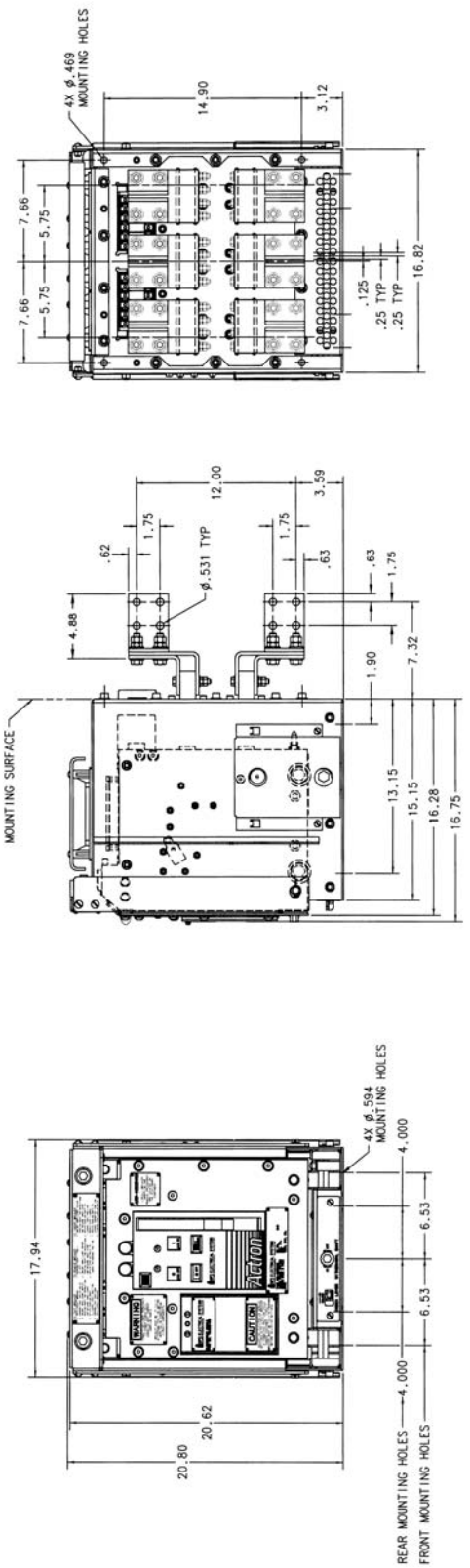



FIGURE 9. Outline and mounting dimensions for circuit breaker type ACB-1606HRC.



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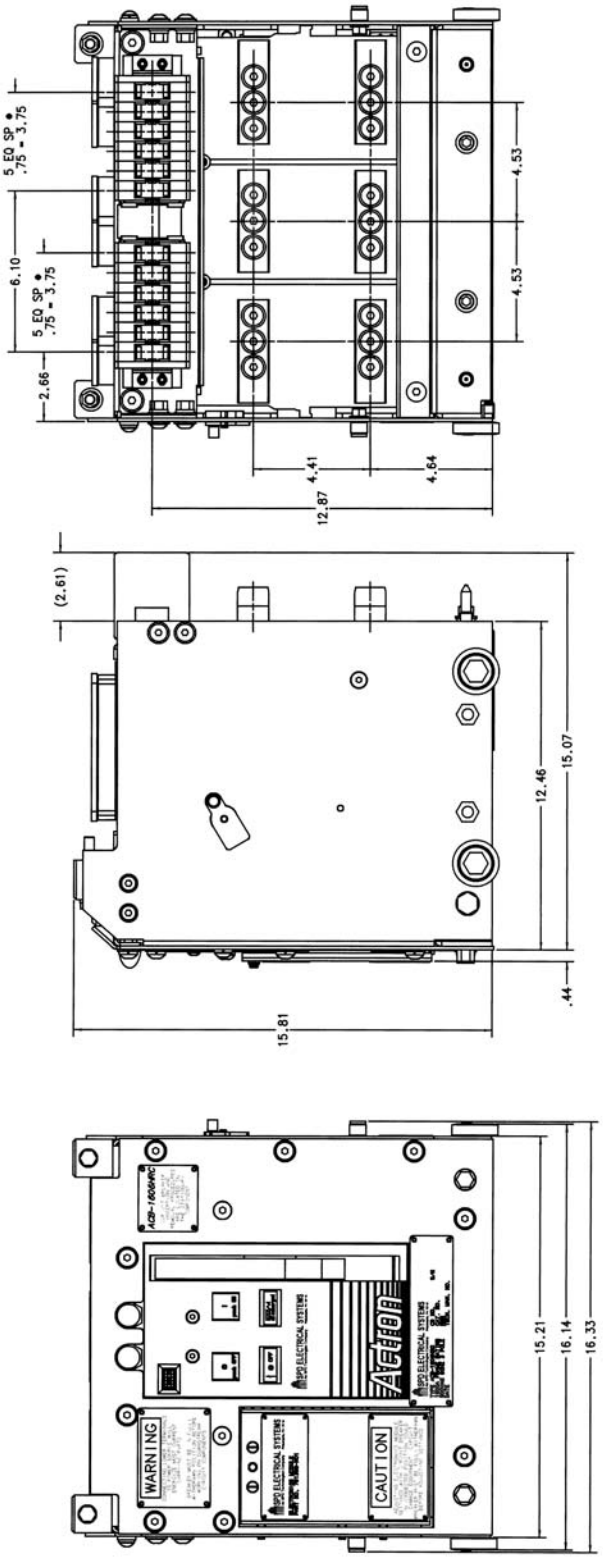



FIGURE 10. Outline and mounting dimensions for circuit breaker type ACB-1606HRC (removable element for).



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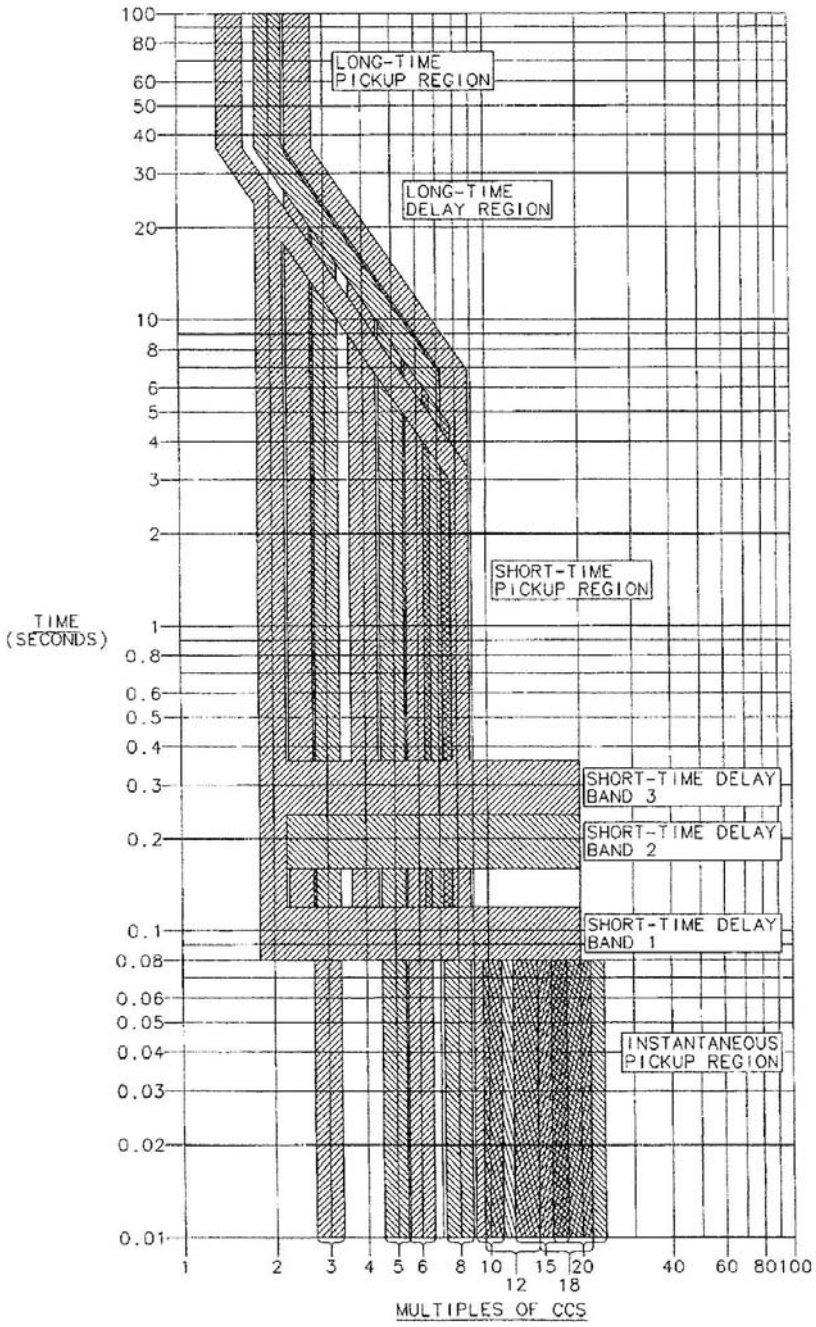



FIGURE 11. Time-current characteristic curve for circuit breaker type ACB-1606HRC.



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| ADJUSTABLE RATINGS AND CONTINUOUS CURRENT SETTINGS (CCS) | | | | | |
|---|---|---|---|---|---|
| 600A | 800A | 1000A | 1200A | 1400A | 1600A |

| ADJUSTABLE LONG-TIME DELAY PICKUP SETTINGS (IN MULTIPLES OF CONTINUOUS CURRENT SETTINGS) | | |
|---|---|---|
| 1.5X | 2.0X | 2.5X |

| ADJUSTABLE SHORT-TIME DELAY PICKUP SETTINGS (IN MULTIPLES OF CONTINUOUS CURRENT SETTINGS) | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2.0X | 2.5X | 3.0X | 4.0X | 5.0X | 6.0X | 7.0X | 8.0X |

| ADJUSTABLE INSTANTANEOUS PICKUP SETTINGS (IN MULTIPLES OF 1600 AMPERE FRAME RATING) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 3X | 5X | 6X | 8X | 10X | 12X | 15X | 18X | 20X |
| 4800 | 8000 | 9600 | 12800 | 16000 | 19200 | 24000 | 28800 | 32000 |

| ADJUSTABLE SHORT-TIME DELAY SETTINGS | |
|---|---|
| SHORT-TIME DELAY BAND | DELAY (SECONDS) |
| 1 | .080-.120 |
| 2 | .160-.240 |
| 3 | .240-.360 |

PER MIL-DTL-17587

LONG-TIME AND SHORT-TIME PICKUP CURRENTS ARE ACCURATE TO ±10%. INSTANTANEOUS PICKUP CURRENTS ARE ACCURATE TO:

±10% FOR PICKUPS ≤ 10X FRAME RATING
±20% FOR PICKUPS > 10X FRAME RATING

LONG-TIME AND SHORT-TIME CALIBRATION ARE GUARANTEED OVER TEMPERATURE RANGE DEFINED BY SWITCHGEAR AMBIENT OF 10°C TO 50°C.

INSTANTANEOUS TRIP POINTS ARE GUARANTEED AT 25°C ONLY. CONSULT FACTORY FOR ACCURACY AT HIGHER TEMPERATURE.

TRIP TIMES ARE FROM INITIATION OF SYMMETRICAL SINUSOIDAL CURRENT TO EXTINGUISHING OF ARC AT RATED VOLTAGE.

FIGURE 11. Time-current characteristic curve for circuit breaker type ACB-1606HRC - Continued.



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